Relation oversampling in the factor-base log precompute — frozen run run-20261111T0602Z

Protocol: `research/logs_oversampling_20261111/protocol.json`, frozen 2026-11-11T05:54Z before this data, sha256 `aa928f27f9be1dff19e1d62ee79a148ac14c06d56c0dd80193bef62ae1153076`. All numbers `[pcc:unverified]`.

Outcome and checks

- 4 cells × 30 seeds × 6 arms = 720 rows requested, 720 completed, **0 failed verification** (each arm re-checked `[log]G == point` for every column).
- **Decision (protocol rule): H0 for every f in every cell.** Every f ∈ {0.1, 0.25, 0.5} has a mean paired ΔC > 0 with the 95% bootstrap interval entirely above 0 at $\bar{\lambda}$, and ΔC > 0 at λ = 0, $\bar{\lambda}$/10 and 10$\bar{\lambda}$. The `ic logs` reference arm (`logs_default`) is also costlier than f = 0 in every cell.
- Mechanism: the extra relations cost trials that the filter/LA cannot win back. LA is 0.8–15.7% of f0 wall time. In only two cell–arm pairs did LA work fall on average (C3 f0.1: −17.9 row_ops; C4 f0.1: −4.2); for oversampling to pay there, a row operation would have to cost 0.61 and 2.6 trials respectively, about 98× and 14,000× the measured $\bar{\lambda}$ (`full/break_even.json`).
- Consistent with the 5-seed pilot (run-20261111T0451Z, disjoint seeds).

What ran

- `research/logs_oversampling_20261111/full/run_full.sh` under bench record `benchmarks/records/20261111T0602Z-logs-oversampling-full.json`; release `ic` rebuilt at the frozen commit (branch `aburan28/kind-volta-4zo2ul`, after merging main); `ic oversample --seeds 30 --seed 0x4b6f626c69747a64 --fractions 0,0.1,0.25,0.5 --aa-control --logs-arm`, `RAYON_NUM_THREADS=1`. Wall 2 min 45 s.
- Host: shared cloud container, Intel Xeon, 4 CPUs, isolation **unstated**. Owner-requested local route (isolab-eks unavailable: no AWS identity); S3 resume checkpoint waived by the owner. The primary metric is in counted units (trials, row_ops), exact; wall values are indicative only.
- Cells: C1 deg 23 m=2 factor-index 0 (44 cols); C2 deg 37 m=3 `ic search` selection (16 cols); C3 deg 37 m=3 subgroup_orbits 7400, CADO filter (104 cols); C4 = C3 with legacy filter.
- Protocol gaps recorded at freeze: no research-kb receipt (not reachable), no independent reviewer.

Results (derived by `full/analyze.py` from `full/rows/*.jsonl`)

C(f) = trials + λ·row_ops; $\bar{\lambda}$ = median over f0 arms of (LA s / row_op) ÷ (collection s / trial). ΔX = arm − f0 on the same seed's pool. CI: paired bootstrap over 30 seeds, 10,000 resamples.

cell	cols	failed	$\bar{\lambda}$ (trials/row-op)	mean C(f0) at $\bar{\lambda}$	f0 core dim range	LA share of f0 wall	A/A max |ΔLA| (ms)
C1	44	0/180	0.104	312.5	0–0	15.67%	0.558
C2	16	0/180	0.0164	29.5	0–0	9.99%	0.081
C3	104	0/180	0.00627	269.6	0–1	4.87%	1.008
C4	104	0/180	0.000181	259.3	7–14	0.83%	0.209

cell	arm	Δtrials	Δrow_ops	ΔC at $\bar{\lambda}$, 95% CI	ΔC at 10$\bar{\lambda}$	Δwall (ms)
C1	f0.1	12.1	8.2	12.9 [11.3, 14.8]	20.6	0.40
C1	f0.25	27.6	20.8	29.7 [27.3, 32.3]	49.2	0.86
C1	f0.5	57.1	39.8	61.2 [57.5, 64.9]	98.4	1.66
C1	logs_default	22.9	17.6	24.8 [18.2, 31.6]	41.2	0.75
C2	f0.1	2.8	37.1	3.4 [3.1, 3.8]	8.9	0.10
C2	f0.25	5.5	63.5	6.5 [6.0, 7.1]	15.9	0.19
C2	f0.5	11.4	113.6	13.2 [12.6, 13.9]	30.0	0.39
C2	logs_default	37.1	279.7	41.7 [39.2, 44.0]	82.9	1.26
C3	f0.1	11.0	−17.9	10.9 [10.5, 11.2]	9.9	2.04
C3	f0.25	26.0	35.0	26.2 [25.9, 26.6]	28.2	4.86
C3	f0.5	52.0	71.2	52.4 [51.9, 53.0]	56.5	9.68
C3	logs_default	33.0	55.6	33.3 [27.2, 39.7]	36.5	6.20
C4	f0.1	11.0	−4.2	11.0 [10.9, 11.1]	11.0	1.99

C4	f0.25	26.0	216.1	26.0 [25.9, 26.2]	26.4	4.72
C4	f0.5	52.0	77.1	52.0 [51.8, 52.2]	52.1	9.49
C4	logs_default	33.0	254.9	33.0 [26.9, 39.4]	33.5	6.05

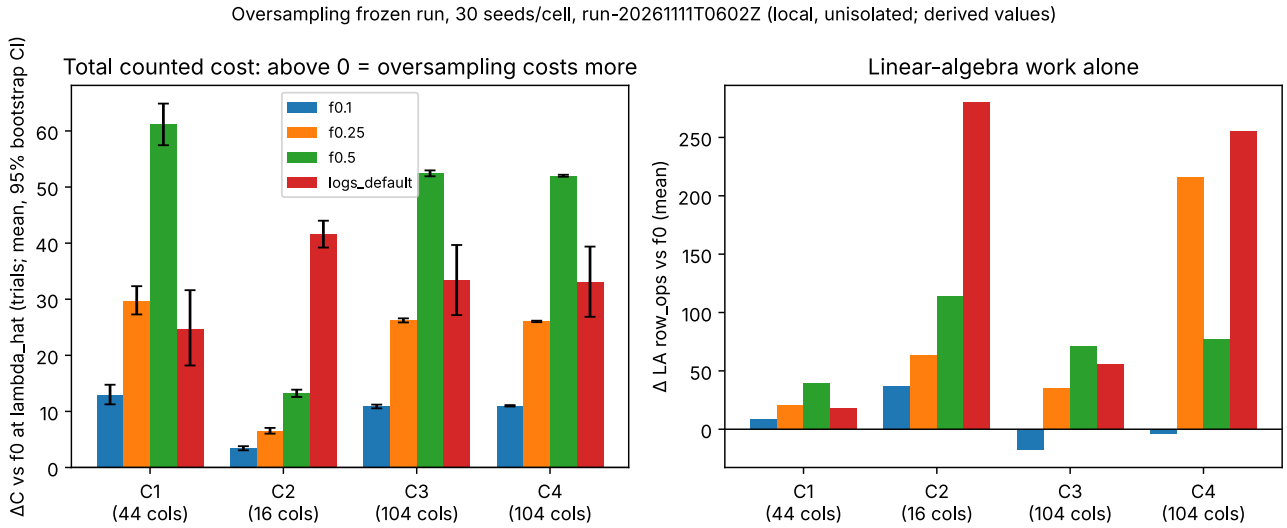


Figure 1: Paired differences vs f0 per cell, 30 seeds each, run-20261111T0602Z. Left: total counted cost at $\hat{\lambda}$ (mean, 95% bootstrap CI). Right: LA row operations (mean). Derived values.

Scope

This measures `ic logs` precompute on these four small Koblitz cells, where filtering leaves a Krylov core of 0–14. It says nothing about cells with large cores (hundreds of columns or more), where LA could dominate. No improvement over the index-calculus baseline was found; no leaderboard row.

Raw records

`research/logs_oversampling_20261111/full/rows/C{1..4}.jsonl` and `.out.json`; summary `full/full_summary.json`. Hashed in the bench record; not uploaded to S3 (no AWS identity).